

CHKDSK Loop Fix

Issue: CHKDSK runs after every reboot. Volume is flagged dirty and won't clear.

Decision Tree

```
CHKDSK runs every reboot
■■■ Check dirty bit: fsutil dirty query C:
■   ■■■ "Volume - C: is Dirty"
■   ■   ■■■ Run chkdsk C: /f /r → Did it run on reboot?
■   ■   ■   ■■■ YES, completed → Check dirty bit again
■   ■   ■   ■   ■■■ Still dirty → FIX B (persistent dirty bit)
■   ■   ■   ■   ■■■ Clean → Resolved. Monitor for recurrence.
■   ■   ■   ■■■ NO, skipped
■   ■   ■       → chkntfs C: → auto-check disabled?
■   ■   ■       ■■■ YES → chkntfs /d (reset default) → reboot
■   ■   ■       ■■■ NO → chkdsk C: /f /r /x (force dismount)
■   ■   ■■■ Cannot run chkdsk → Schedule: chkdsk C: /f /r → reboot
■   ■■■ "Volume - C: is NOT Dirty"
■       ■■■ Check scheduled tasks and chkntfs /t for misconfiguration
```

FIX A: Standard CHKDSK Repair

```
chkdsk C: /f /r /x
```

If volume is in use, type Y to schedule on next reboot. Let it complete fully (1-4 hours).

Verify after:

```
fsutil dirty query C:
```

FIX B: Persistent Dirty Bit

1. Check for drivers re-dirtying the volume:

```
wevtutil qe System /q:"*[System[Provider[@Name='Ntfs'] and (EventID=55 or EventID=137)]" /c:10 /f:0
```

2. Run CHKDSK from recovery environment (volume fully dismounted):

- Boot from Windows installation media or WinPE.
- Open command prompt.
- Run **chkdsk C: /f /r**.

3. If CHKDSK finds bad sectors, check drive health:

```
Get-PhysicalDisk | Select-Object MediaType, HealthStatus, OperationalStatus
```

If HealthStatus is "Warning" or "Unhealthy" → back up data immediately and replace the drive.

Temporary Band-Aid

```
:: Stop the loop while you investigate (not a fix)
chkntfs /x C:
```

```
:: Re-enable later
chkntfs /d
```

Prevention

- Monitor drive health with SMART data.
- Ensure clean shutdown procedures — dirty shutdowns are the #1 cause.
- If recurring on the same machine, the drive is likely degrading. Replace proactively.